MarkIn - Smart Attendance Management System

Synopsis for Project

Bachelor Of Technology

in

Computer Science & Engineering

Submitted by

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Abstract

This project presents a Smart Attendance Management System that modernizes academic attendance through a secure, contactless, and web-based platform. The system uses a React frontend, an Express and MongoDB backend, and a FastAPI AI service to support class creation, student enrollment, face-profile registration, and AI-assisted attendance verification. Teachers can manage classrooms and finalize attendance records, while students can join classes and track their attendance through dedicated dashboards.

The platform is designed around face-based recognition with support for detection, recognition, liveness verification, and anti-spoofing, making the attendance workflow more reliable and transparent than traditional manual methods. Along with the implemented features, the system is also planned to expand with GEI-based attendance, a community section, and an advanced analytics section in future development. This makes the project a scalable foundation for smart attendance management in educational institutions.

1. Introduction

Attendance management is one of the most important academic and administrative activities in educational institutions, as it directly affects student discipline, participation, internal evaluation, and eligibility criteria. Maintaining accurate attendance records is essential not only for institutional transparency but also for helping teachers and students monitor classroom engagement over time. However, in many institutions, attendance is still recorded using manual or semi-digital methods that are time-consuming, error-prone, and difficult to manage efficiently, especially in classes with large student strength.

Traditional approaches such as roll calls, paper registers, and signature sheets consume valuable teaching time and remain vulnerable to proxy attendance and human mistakes. More recent methods such as RFID cards, QR-code scanning, and fingerprint-based devices improve automation to some extent, but they still introduce practical limitations. Cards and QR codes can be shared or misused, biometric devices require dedicated hardware, and many such systems do not provide a flexible, integrated academic workflow for both teachers and students. These limitations highlight the need for a smarter, more scalable, and more transparent attendance solution.

This project introduces a **Smart Attendance Management System** that combines web-based classroom management with AI-assisted attendance verification. The system provides separate workspaces for teachers and students, allowing teachers to create classes, generate join codes and links, manage rosters, and conduct attendance sessions, while students can register, join classrooms, enroll their face profiles, and track attendance through dedicated dashboards. The platform is implemented using a multi-service architecture with a **React frontend**, **Express and MongoDB backend**, and a **FastAPI-based AI service**, creating a modular and extensible foundation for smart academic management.

The attendance workflow is centered around face-profile enrollment, classroom image capture, automated recognition, and teacher-reviewed finalization. The AI layer is structured to support face detection, recognition, liveness verification, and anti-spoofing, while the backend manages authentication, classrooms, memberships, attendance records, and dashboard analytics. A teacher-in-the-loop verification process ensures that suggested matches, low-confidence cases, and absent students can be reviewed before attendance is finalized, improving reliability and maintaining human oversight in the decision-making process.

In addition to the features currently implemented, the system is designed for future enhancement. Planned extensions include **GEI-based attendance**, a **community section** for academic interaction and communication, and an **advanced analytics section** for deeper attendance insights, reporting, and decision support. Thus, the proposed system not only addresses present attendance management challenges but also provides a scalable base for future smart campus solutions.

2. Motivation

Attendance management is a fundamental academic and administrative process in educational institutions, as it directly affects student discipline, participation tracking, academic review, and eligibility requirements. Despite its importance, many institutions still rely on attendance methods that are either manual or fragmented across different tools. These approaches increase classroom overhead, reduce teaching efficiency, and make it difficult to maintain accurate and transparent records over time. This challenge creates the need for a smarter and more integrated attendance platform.

Traditional practices such as roll calls, paper registers, and signature sheets are time-consuming and highly vulnerable to proxy attendance and human error. Even when institutions adopt partially digital alternatives, many systems focus only on the act of marking attendance and ignore the larger workflow surrounding class creation, roster management, student onboarding, and long-term record monitoring. As a result, teachers often have to manage attendance, class coordination, and reporting separately, which increases effort and reduces overall efficiency.

Other automated approaches such as RFID cards, QR-based systems, and hardware-dependent biometric devices improve convenience only to a limited extent. Cards and QR codes can be shared or misused, while device-based solutions increase deployment cost and hardware dependence. Many such systems also fail to provide a centralized platform where students and teachers can interact through dedicated workspaces, track attendance history, and maintain class-level academic visibility. This motivates the development of a web-based solution that is both practical and scalable for real academic environments.

The present project is motivated by the need to combine attendance automation with structured classroom management and human-verified reliability. In this system, teachers can create classes, generate join codes and links, manage rosters, and conduct attendance sessions, while students can join classrooms, complete face-profile enrollment, and monitor their attendance from dedicated dashboards. Instead of relying on blind automation, the platform follows an AI-assisted and teacher-reviewed attendance workflow, which improves trust, control, and usability.

Another major motivation behind this project is to build a modular and future-ready academic platform rather than a one-purpose attendance tool. The current architecture already supports web-based attendance, dashboard insights, and class management, while also creating a strong foundation for future enhancements such as GEI-based attendance, a community section for academic interaction, and a dedicated analytics section for deeper reporting and decision support. Thus, the project is driven by the vision of transforming attendance from a repetitive administrative task into an intelligent, extensible, and institution-friendly digital system.

3. Objective

The primary objective of the **MarkIn - Smart Attendance Management System** is to design and develop a secure, contactless, and automated attendance solution using modern Artificial Intelligence, Deep Learning, and Computer Vision techniques. The system aims to overcome the limitations of traditional attendance methods by ensuring accurate identity verification and preventing proxy attendance.

The specific objectives of the project are as follows:

- To develop a secure, web-based smart attendance management system for academic use.
- To automate student attendance marking using face enrollment and face recognition techniques.
- To reduce proxy attendance through AI-based identity verification and review-supported attendance processing.
- To provide separate role-based access and dashboards for students and teachers.
- To enable class creation, class joining, attendance monitoring, and attendance finalization through a unified platform.
- To maintain centralized attendance records for tracking, management, and academic reporting.
- To design the system as a scalable and practical solution for future enhancements and real-world deployment.

Overall, the objective of this project is to build a **modern, intelligent, and proxy-resistant attendance system** that improves efficiency, accuracy, and transparency in academic environments while remaining practical for real-world deployment.

4. Problem Statement

- Traditional attendance methods such as manual registers are slow, repetitive, and prone to human error.
- Existing systems often fail to verify the actual identity of the student, which allows proxy attendance.
- Hardware-dependent solutions like fingerprint scanners and RFID cards increase cost, maintenance, and operational complexity.
- Many attendance methods do not provide a secure, contactless, and convenient experience for daily classroom use.
- Attendance records are often not centralized properly, making monitoring, reporting, and academic analysis difficult.
- Teachers need a faster and more reliable way to manage classes, verify attendance, and maintain accurate records.
- Therefore, there is a need for an intelligent web-based attendance system that uses AI-assisted face recognition to improve accuracy, security, transparency, and efficiency in academic institutions.

5. Literature Survey

Attendance management systems have evolved significantly over time, transitioning from manual procedures to automated and intelligent systems using biometric and artificial intelligence-based techniques. This section reviews existing attendance approaches, deep learning-based face recognition methods, liveness detection techniques, and real-time face detection frameworks relevant to the proposed Smart Attendance Management System.

5.1 Traditional Attendance Systems

Conventional attendance methods such as manual roll calls, paper registers, and signature sheets have been widely used in academic institutions for decades. While simple to implement, these methods are highly time-consuming, prone to human error, and vulnerable to proxy attendance. In large classrooms, manual systems lead to significant wastage of instructional time and unreliable record maintenance.

To address these issues, token-based systems such as RFID cards, QR codes, and ID-based attendance were introduced. However, as discussed by Jain et al. [4], token-based approaches ("what you have") do not guarantee identity authenticity because cards and QR codes can be easily shared or misused. These systems also fail to verify the physical presence of the student.

5.2 Biometric-Based Attendance Systems

Biometric attendance systems use physiological traits such as fingerprints, iris scans, or palmprints for identity verification. Jain et al. [4] provide a comprehensive overview of biometric systems, highlighting their advantages over token-based methods. While biometric systems improve identity assurance, fingerprint-based attendance systems suffer from hygiene concerns, require dedicated hardware, and remain vulnerable to spoofing using artificial fingerprints.

Additionally, biometric hardware installations are expensive and unsuitable for large-scale classroom environments, making them less practical for continuous academic use.

5.3 Classical Face Recognition Techniques

Before the advent of deep learning, face recognition relied on handcrafted feature extraction techniques such as Eigenfaces (PCA), Fisherfaces (LDA), Local Binary Patterns (LBP), and HOG-based classifiers. These methods, reviewed in the YOLO-based face recognition analysis by Sun [3], are computationally efficient but highly sensitive to illumination changes, pose variations, facial expressions, and background noise.

Due to their limited robustness, classical methods are unsuitable for real-world attendance systems that operate under uncontrolled classroom conditions.

5.4 Deep Learning-Based Face Recognition

Deep learning revolutionized face recognition by enabling end-to-end learning of facial features directly from data. One of the most influential works in this domain is **FaceNet**, proposed by Schroff et al. [2]. FaceNet introduces a unified embedding framework that maps facial images into a compact Euclidean space, where distances directly represent identity similarity.

By training a deep convolutional neural network using triplet loss, FaceNet ensures that embeddings of the same person are close together while embeddings of different individuals are far apart. This embedding-based approach enables scalable, accurate, and efficient face recognition using simple distance metrics.

Recent attendance systems, such as the work by Saile et al. [5], successfully integrate FaceNet with face detectors like MTCNN and similarity search libraries like FAISS to automate attendance marking. These systems demonstrate the feasibility of deep learning-based face recognition for academic environments.

5.5 Face Detection for Real-Time Systems

Accurate face detection is a critical prerequisite for face recognition. Sun [3] analyzes the use of YOLO (You Only Look Once)-based object detection models for real-time face detection. Unlike multi-stage detectors, YOLO performs detection in a single forward pass, achieving high speed and real-time performance.

YOLO-based detectors are especially useful in scenarios requiring multi-face detection, such as classroom entry points or surveillance-based attendance systems. While the current prototype may rely on lighter detectors such as MediaPipe or MTCNN, YOLO-based approaches provide a strong foundation for future scalability and multi-face attendance automation.

5.6 Liveness Detection and Anti-Spoofing Techniques

Although deep learning–based face recognition achieves high accuracy, it is vulnerable to spoofing attacks using printed photographs, mobile screens, or replayed videos. To address this issue, liveness detection techniques are essential.

Soukupová and Čech [1] propose a real-time eye-blink detection method using facial landmarks and the Eye Aspect Ratio (EAR). Their approach demonstrates that natural blink patterns can be reliably detected using standard RGB webcams without specialized hardware. This landmark-based liveness detection method is lightweight and suitable for real-time systems.

Texture-based and CNN-based anti-spoofing techniques further enhance security. Boulkenafet et al. [6] demonstrate that deep learning models can distinguish real facial textures from printed or digital spoof artifacts. These methods significantly reduce false acceptance rates in face recognition systems.

5.7 Research Gaps Identified

Despite significant progress, existing systems exhibit several limitations:

- Many attendance systems rely solely on face recognition without robust liveness detection, making them vulnerable to spoofing attacks.
- Token-based and hardware-dependent systems fail to ensure identity authenticity and physical presence.
- YOLO-based detectors offer speed but lack built-in identity verification, while FaceNet provides embeddings but lacks inherent spoof resistance.
- Few systems integrate multi-frame liveness detection, deep learning face recognition, and secure session-level validation into a single unified platform.

These gaps highlight the need for a **contactless, spoof-resistant, AI-driven attendance system** that combines deep learning-based face recognition, landmark-based liveness detection, and secure web-based automation-motivating the proposed Smart Attendance Management System.

6. Methodology

This section explains the existing attendance methodology based on face detection and recognition, followed by the proposed Smart Attendance Management System.

6.1 Existing Methodology

Earlier face-recognition–based attendance systems focused on automating attendance using **face detection and identity matching**, eliminating the need for manual roll calls or roll numbers. In such systems, student faces are captured from images or live webcam input, and attendance is automatically recorded in a CSV file for later use [5].

When a new image or webcam frame is received, the system first performs image resizing and enhancement, followed by **face detection using MTCNN**. The detected face is then converted into a **128-dimensional facial embedding** using a deep learning model inspired by **FaceNet**, which learns discriminative facial representations through CNN architectures and triplet loss [2]. These embeddings are compared with previously stored embeddings using **FAISS**, which enables fast similarity matching. The closest matching embedding is selected as the recognized student, and attendance is finally stored in a CSV file.

MTCNN performs face detection using a **three-stage pipeline** consisting of P-Net for generating candidate face regions, R-Net for removing false positives, and O-Net for final face detection along with five facial landmarks. FAISS efficiently manages embedding storage and similarity search, while **Flask** is used to provide a simple web interface for camera streaming, face capture, recognition, and attendance logging [5].

Although this approach automates attendance, it **does not include liveness detection or session-level security**, making it vulnerable to spoofing attacks such as printed photos or screen replays.

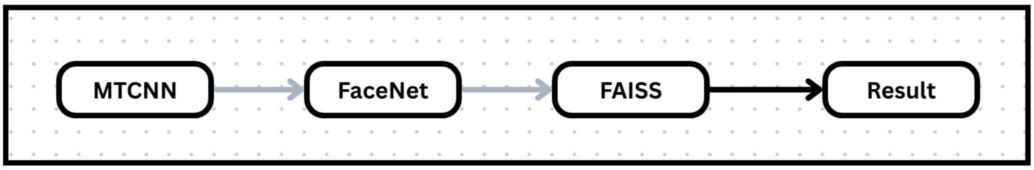

Figure 6.1.1: Existing Face Detection and Recognition Attendance Workflow

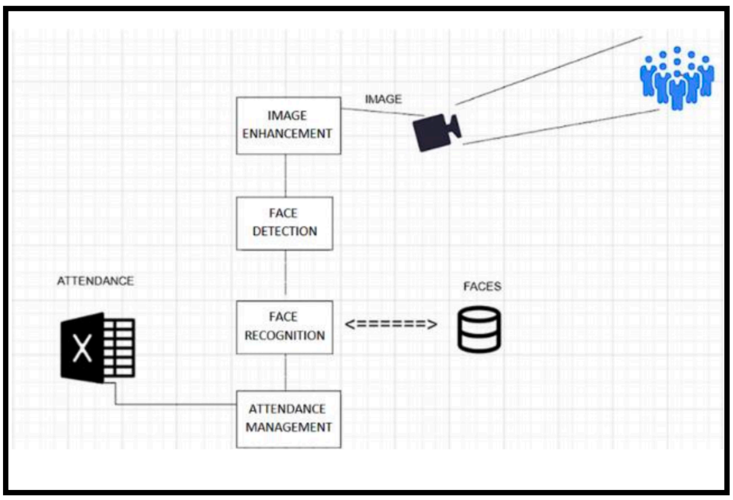

Figure 6.1.2: Visual Workflow

6.2 Proposed Methodology

The proposed methodology for the MarkIn - Smart Attendance Management System is based on a modular, multi-service, and workflow-oriented approach in which attendance management is carried out through the coordinated operation of a web frontend, backend server, AI service, and centralized database. The methodology is designed to ensure secure access, reliable identity verification, contactless attendance marking, teacher-controlled review, and centralized record maintenance. The system is structured to support real academic usage while also remaining scalable for future deployment with stronger production-grade AI models.

- **Requirement Analysis and Problem Definition**
  The first stage involves identifying the limitations of traditional attendance systems such as manual errors, proxy attendance, slow record handling, and lack of centralized monitoring. Based on this analysis, the major system requirements are defined, including role-based access, classroom management, face enrollment, AI-assisted attendance verification, review-driven finalization, and attendance analytics.

- **System Architecture Design**
  The system is designed using four major layers: frontend, backend, AI service, and database. The frontend provides the user interface for students and teachers. The backend handles authentication, classroom workflows, business logic, and communication with the database and AI service. The AI service is responsible for face enrollment, recognition, liveness verification, and anti-spoof analysis. The database stores user details, classrooms, face-profile summaries, and finalized attendance records. This separation improves maintainability, scalability, and security.

- **User Management and Classroom Workflow**
  Role-based access control is used so that students and teachers operate within their own secure modules. Teachers can create classrooms, generate join codes and links, and manage class rosters. Students can register, log in, join classes, and access their attendance dashboards. This ensures that attendance processing is always linked to an authorized classroom roster rather than being performed on unrestricted identity data.

- **Face Enrollment Methodology**
  Before a student can participate reliably in face-based attendance, the student must enroll a face profile by uploading multiple face images. These images are used to generate reference facial embeddings, and an average embedding is created to represent the enrolled identity. The enrolled profile is then stored and associated with the student account. This profile becomes reusable across all joined classes, reducing repeated enrollment effort and improving attendance workflow efficiency.

- **Teacher-Initiated Attendance Session Creation**
  Attendance is initiated by the teacher from the classroom workspace. The teacher uploads one or more classroom images or captures them directly from the camera. Each attendance activity is treated as a session with a unique session identifier. The teacher's class roster is sent along with the captured images so that recognition is restricted to authorized students of that class only. This session-based design improves control, traceability, and reviewability.

- **AI-Based Face Recognition and Attendance Processing**
  The AI service processes the attendance session in multiple stages. First, face detection is performed to locate faces in the captured classroom image. Then tracking logic is used to organize detected faces into candidate face tracks. For each detected face, an embedding is generated and compared with the enrolled facial embeddings of students present in the class roster. Similarity matching is then applied to determine the best identity match. Based on the similarity score and quality conditions, the system categorizes results into three groups: automatically suggested present students, low-confidence matches requiring review, and unknown detections. Students from the roster who are not recognized in the session are initially treated as absent candidates.

- **Liveness Detection and Anti-Spoof Protection**
  To reduce the possibility of spoofing using photographs, screens, or prerecorded media, the methodology includes liveness and anti-spoof verification as a separate AI verification layer. This stage checks whether the detected face appears to be live by using signals such as blink detection and head movement analysis, along with passive anti-spoof scoring. These checks strengthen identity reliability and improve resistance against proxy attendance attempts.

- **Teacher Review and Attendance Finalization**
  Instead of allowing the AI system to mark attendance permanently without supervision, the proposed methodology follows a review-first approach. The teacher is shown suggested present students, low-confidence matches, absent candidates, and unknown detections. The teacher can accept, reject, or manually correct these results before final submission. This step ensures that the final attendance record remains academically reliable, transparent, and practical for real-world classroom use.

- **Attendance Storage, Monitoring, and Reporting**
  After teacher approval, the final attendance data is stored in the centralized database. These records are then used to generate student and teacher dashboards, attendance summaries, class-wise statistics, recent session history, and exportable attendance reports. Centralized record management improves long-term monitoring, academic analysis, and administrative accessibility.

- **Testing, Evaluation, and Future Scalability**
  The system is tested across individual modules as well as through end-to-end integration. Evaluation includes correctness of recognition flow, usability of the review process, response behavior of the web system, and consistency of stored records. The methodology is intentionally designed to remain extensible so that future enhancements such as real-time multi-face recognition, strict time-window control, cloud deployment, IoT kiosk support, and ERP/LMS integration can be incorporated without major restructuring.

Model Working and Processing Pipeline

The working of the model in this system follows a structured pipeline that begins with face enrollment and ends with teacher-approved attendance storage. The intended model stack consists of face detection, face tracking, face recognition, liveness verification, and anti-spoofing. In architectural terms, the system is prepared for models such as SCRFD for detection, ByteTrack for tracking, ArcFace for recognition, MediaPipe Face Landmarker for liveness cues, and MiniFASNetV2 for anti-spoofing. However, in the current implementation, the end-to-end AI pipeline is fully wired but the inference logic runs in a simulated deterministic mode, which means the workflow is real while the production model weights can be integrated later.

During face enrollment, multiple student images are converted into reference embeddings. These embeddings are mathematically averaged to form a stable identity profile for the student. During attendance processing, each detected classroom face is converted into a capture embedding, and cosine-similarity matching is used to compare it against the roster-specific enrolled profiles. If the similarity score crosses the recognition threshold and no warning flags are present, the student is marked as a suggested present case. If the similarity is moderate or quality conditions are weak, the student is placed in the manual-review queue. If no reliable match is found, the face is treated as unknown. This design allows the system to balance automation with reliability.

The complete operational flow of the system can therefore be summarized as follows: student registration and class joining, face enrollment, teacher-initiated attendance session, classroom image capture, AI-based face detection and recognition, liveness and anti-spoof assessment, teacher review of results, final attendance submission, and centralized storage for dashboards and reporting. This methodology ensures that the system is not only intelligent and automated, but also practical, review-aware, and suitable for academic deployment.

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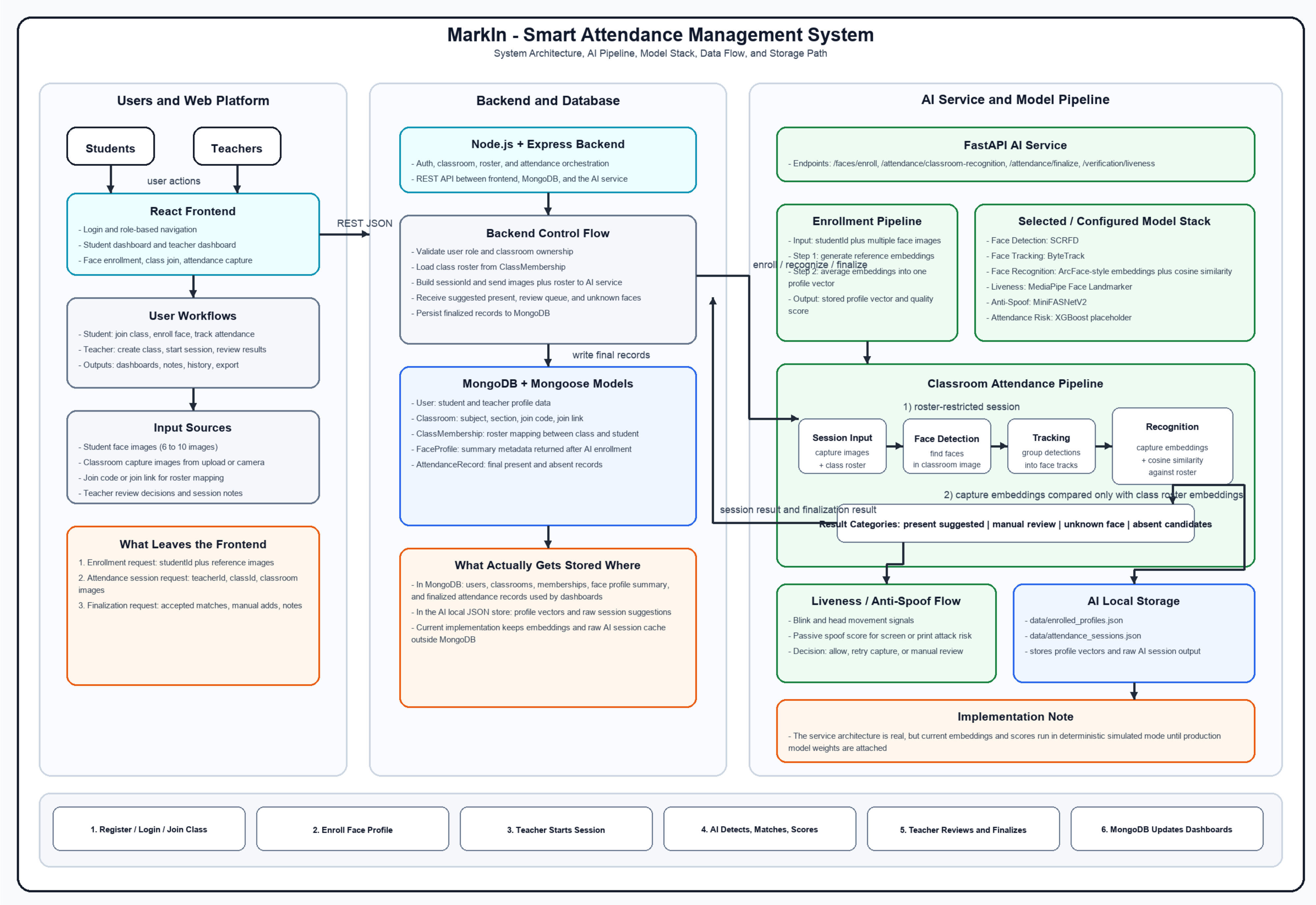
MarkIn - Smart Attendance Management System
System Architecture, AI Pipeline, Model Stack, Data Flow, and Storage Path
Users and Web Platform
Students
Teachers
user actions
React Frontend
- Login and role-based navigation
- Student dashboard and teacher dashboard
- Face enrollment, class join, attendance capture
User Workflows
- Student: join class, enroll face, track attendance
- Teacher: create class, start session, review results
- Outputs: dashboards, notes, history, export
Input Sources
- Student face images (6 to 10 images)
- Classroom capture images from upload or camera
- Join code or join link for roster mapping
- Teacher review decisions and session notes
What Leaves the Frontend
1. Enrollment request: studentId plus reference images
2. Attendance session request: teacherId, classId, classroom images
3. Finalization request: accepted matches, manual adds, notes
REST JSON
Backend and Database
Node.js + Express Backend
- Auth, classroom, roster, and attendance orchestration
- REST API between frontend, MongoDB, and the AI service
Backend Control Flow
- Validate user role and classroom ownership
- Load class roster from ClassMembership
- Build sessionId and send images plus roster to AI service
- Receive suggested present, review queue, and unknown faces
- Persist finalized records to MongoDB
write final records
MongoDB + Mongoose Models
- User: student and teacher profile data
- Classroom: subject, section, join code, join link
- ClassMembership: roster mapping between class and student
- FaceProfile: summary metadata returned after AI enrollment
- AttendanceRecord: final present and absent records
What Actually Gets Stored Where
- In MongoDB: users, classrooms, memberships, face profile summary, and finalized attendance records used by dashboards
- In the AI local JSON store: profile vectors and raw session suggestions
- Current implementation keeps embeddings and raw AI session cache outside MongoDB
enroll / recognize / finalize
session result and finalization result
AI Service and Model Pipeline
FastAPI AI Service
- Endpoints: /faces/enroll, /attendance/classroom-recognition, /attendance/finalize, /verification/liveness
Enrollment Pipeline
- Input: studentId plus multiple face images
- Step 1: generate reference embeddings
- Step 2: average embeddings into one profile vector
- Output: stored profile vector and quality score
Selected / Configured Model Stack
- Face Detection: SCRFD
- Face Tracking: ByteTrack
- Face Recognition: ArcFace-style embeddings plus cosine similarity
- Liveness: MediaPipe Face Landmarker
- Anti-Spoof: MiniFASNetV2
- Attendance Risk: XGBoost placeholder
Classroom Attendance Pipeline
1) roster-restricted session
Session Input
capture images
+ class roster
Face Detection
find faces
in classroom image
Tracking
group detections
into face tracks
Recognition
capture embeddings
+ cosine similarity
against roster
2) capture embeddings compared only with class roster embeddings
Result Categories: present suggested | manual review | unknown face | absent candidates
Liveness / Anti-Spoof Flow
- Blink and head movement signals
- Passive spoof score for screen or print attack risk
- Decision: allow, retry capture, or manual review
AI Local Storage
- data/enrolled_profiles.json
- data/attendance_sessions.json
- stores profile vectors and raw AI session output
Implementation Note
- The service architecture is real, but current embeddings and scores run in deterministic simulated mode until production model weights are attached
1. Register / Login / Join Class
2. Enroll Face Profile
3. Teacher Starts Session
4. AI Detects, Matches, Scores
5. Teacher Reviews and Finalizes
6. MongoDB Updates Dashboards

7. Expected Results

- A functional web-based attendance management system for students and teachers will be developed.
- Students will be able to enroll their face profiles and join classes through the platform.
- Teachers will be able to create classes, manage rosters, and conduct attendance sessions digitally.
- The system will support AI-assisted face-based attendance verification for faster and more accurate attendance marking.
- Attendance records will be stored in a centralized manner and can be viewed, monitored, and exported when required.
- The platform will reduce manual effort, improve transparency, and minimize the chances of proxy attendance.
- The system will provide a scalable foundation for future improvements such as advanced liveness detection, cloud deployment, and institutional integration.

8. Plan of Work

Month-wise Plan of Work

December

- Identification of problem and requirement analysis
- Study of existing attendance systems and their limitations
- Literature survey on AI-based attendance, face recognition, and liveness detection
- Finalization of project objectives, modules, and user roles
- Preparation and submission of initial project synopsis

January

- Design of overall system architecture and workflow
- Planning of frontend, backend, AI service, and database integration
- Study of face enrollment, facial embeddings, and similarity-based matching
- Design of database schema, API flow, and attendance process
- Setup of development environment and project structure

February

- Development of authentication and role-based access for students and teachers
- Implementation of class creation, class joining, and roster management modules
- Development of student and teacher dashboard interfaces
- Implementation of face profile enrollment and image upload workflow
- Backend development for user, classroom, and attendance data handling

March

- Implementation of AI-assisted face recognition and attendance verification pipeline
- Integration of liveness detection and anti-spoofing concepts into the workflow
- Development of attendance session processing, review, and finalization logic
- Storage of attendance records and generation of attendance summaries
- Integration of frontend, backend, and AI service for end-to-end flow

April

- End-to-end system testing and debugging
- Performance evaluation for attendance flow, usability, and reliability
- Refinement of dashboards, reports, and export features
- Preparation of final project report, PPT, and presentation materials
- Final review, documentation, and project submission

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